Gumbel Top-$(K{+}J)$ versus hard Top-K, with the code carried between blocks

Research note

2026-11-13; Section 7 added 2026-11-14.

1 Purpose and summary of conclusions

Three questions: does a Gumbel Top-$(K{+}J)$ bottleneck, trained with a policy gradient, beat hard Top-K; which candidate pool J is best; and how should the temperature τ and the policy coefficient λ be set. All runs use abs-TopK selection (Section 2); runs with separately computed selection scores did not train and were paused (`docs/separate-score-failure.pdf`).

- **Gumbel Top-$(K{+}J)$ does not beat hard Top-K.** All 27 tuned 20000-step cells (9 (K, J) pairs $\times$ 3 sampling variants) end within -0.001 to $+0.018$ nats of hard Top-K at the same K, and 0.05 to 0.26 above hard Top-$(K{+}J)$. On the hard ladder every cell sits at the Top-K end of the interval between hard Top-K and Top-$2K$ (one doubling of K is worth 0.03–0.09 nats).
- **J does not matter.** At each K and variant the final CE over J spans at most 0.011 nats. Only with all 8 bottlenecks sampling is one pool consistently best: the narrowest, $J = K$, by up to 0.01.
- **The policy term only hurts.** At step 2500 the loss rises monotonically with λ for every τ (mean over cells). In the first grid, $\lambda = 10^{-2}$ and 10^{-1} collapsed training to the unigram level (CE ≈ 6), and $\lambda = 1$ did so in a 30-step smoke test; $\lambda = 10^{-3}$ collapsed by step 1000. The selected setting is $\lambda = 0$ in all 27 cells: Gumbel noise on the support, no policy gradient.
- **No crossover later in training.** The largest λ that trains, 10^{-4}, was continued from step 2500 to 20000 at $\tau = 0.5$ and 1.2 in all 27 cell–variant pairs (54 runs), with $3 \cdot 10^{-5}$ in its place where the divergence guard stopped it. No run comes closer than 0.024 nats to hard Top-K at any validation step; the closest end 0.027–0.046 above it, and their margin is flat after step 15000 (Section 7). 16 of the 54 runs at $\lambda = 10^{-4}$ collapse to deterministic sampling, 7 of them after training normally until a step between 8600 and 18000.
- **Smallest temperature.** $\tau = 0.2$ is selected in 23 of 27 cells, $\tau = 0.5$ in the other 4; $\tau = 1.2$ costs 0.04–0.12 nats at step 2500 (mean over cells).
- **Sampling at 3 of 8 bottlenecks** roughly halves the cost of a given λ (mean margin $+0.41$ instead of $+0.80$ at $\tau = 0.5$, $\lambda = 10^{-4}$) and ties sampling at all 8 at $\lambda = 0$. Drawing the 3 bottlenecks once per forward or separately per token gives the same results within 0.01 nats.

2 Setup

Model. 8 blocks of width 768 on TinyStories, an activation bottleneck after every block (`residual_out`, $N = 1536$ coordinates), code carried between blocks (`code_residual: true`), no output norm, magnitude–direction decoupling, batch 96×512, seed 1337. Unlike the K+J notes,

the learning rate is *constant* at $6 \cdot 10^{-4}$ after a 500-step warmup (`configs/gumbel_cr/base.yaml`). The bottleneck at block l keeps the support S_l of K coordinates of the code a_l (abs-TopK: the K largest $|a_{l,i}|$); the hard Top-K baselines train with that deterministic support, $K \in \{16, 32, 64, 128, 256\}$.

Gumbel Top-$(K{+}J)$. In training each bottleneck draws its support from the candidate pool $C = \text{TopIndices}_{K+J}(|a|)$,

$$S = \text{TopK}_K\big(|a|_C/\tau + G_C\big), \qquad G_i \sim \text{Gumbel}(0,1), \tag{1}$$

independently per sequence, position and bottleneck, and the trainer adds the policy term

$$\lambda \textstyle\sum_b w_b \ \text{sg}(\ell_b - \beta_b)\, P_b \tag{2}$$

to the CE: ℓ_b is the CE of sequence b, β_b the mean CE of the other sequences of its micro-batch (leave-one-out baseline), P_b the sequence's summed ordered Plackett–Luce log-probability of its draws, and w_b its weight in the CE. Validation uses the deterministic Top-K. The coordinates kept carry their values unscaled.

Variants. All 8 bottlenecks sample (`gcr_gum_*`); or 3 of the 8, drawn uniformly once per forward (`gcr_gum3f_*`) or once per position (`gcr_gum3t_*`), the other 5 using their deterministic Top-K.

Grid and search. $(K, K{+}J) \in \{16\} \times \{32, 64, 128, 256\} \cup \{32\} \times \{64, 128, 256\} \cup \{64\} \times \{128, 256\}$. At each of the 9 cells and each variant, $\tau \in \{0.2, 0.5, 1.2\}$ (at initialization these put about 9 %, 27 % and 63 % of the drawn support outside Top-K at $K = 32$, pool N) and $\lambda \in \{0, 10^{-5}, 3{\cdot}10^{-5}, 10^{-4}\}$. Search runs stop at step 2500; under the constant learning rate they are the first 2500 steps of the full run, and the setting with the lowest validation CE at step 2500 is resumed from its own checkpoint to step 20000.

3 Hard Top-K

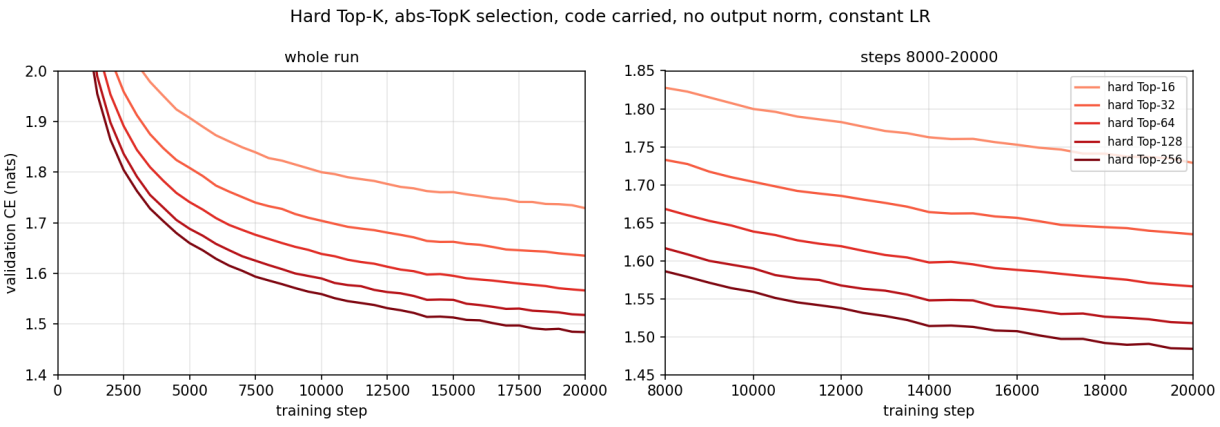


Figure 1: Hard Top-K, $K = 16$ to 256: the whole run, and steps 8000–20000.

Final validation CE: 1.7288, 1.6348, 1.5662, 1.5178, 1.4840 for $K = 16, 32, 64, 128, 256$; each doubling of K lowers the loss by 0.094, 0.069, 0.048 and 0.034 nats (Figure 1).

4 The (τ, λ) search

λ. The loss rises with λ at every τ (Figure 2). The norm of the policy term's gradient, relative to the CE gradient on the same micro-batch, scales with λ: median 0.09, 0.28 and 0.9–1.6 at $\lambda = 10^{-5}, 3 \cdot 10^{-5}, 10^{-4}$ with all 8 bottlenecks sampling, about 0.6 times that with 3 of 8. At

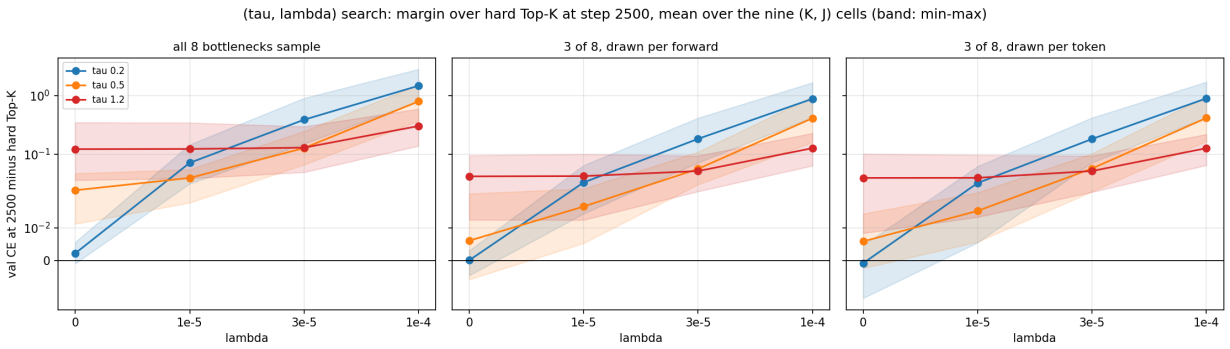


Figure 2: Validation CE at step 2500 minus hard Top-K at step 2500, mean over the 9 cells (band: minimum to maximum over cells), symmetric-log scale.

$\lambda = 10^{-2}$ it reached 10^2–10^3: with the gradient clipped at norm 1, the CE gradient then all but vanishes from the update, the code's scale grows until sampling is deterministic, and the loss returns to the unigram level.

τ**.** At $\lambda = 0$ the noise alone costs nothing at $\tau = 0.2$ (mean margin +0.002, 0.000 and −0.001 for the three variants), little at $\tau = 0.5$ (+0.024 with all 8, +0.006 with 3 of 8), and 0.04–0.12 at $\tau = 1.2$. A larger τ makes the loss less sensitive to λ but adds this cost.

5 Final results

Validation CE after 20000 steps; columns are the candidate pool $K{+}J$, the last column hard Top-K at the same K, the last row hard Top-$(K{+}J)$. Selected (τ, λ): $\lambda = 0$ everywhere, $\tau = 0.2$ except $\tau = 0.5$ at $(16, 16)$ and $(32, 32)$ for 3 of 8 per forward and at $(32, 32)$ and $(64, 64)$ for 3 of 8 per token.

variant	K	$K{+}J = 32$	64	128	256	hard Top-K
all 8 sample	16	1.7381	1.7403	1.7432	1.7436	1.7288
	32	–	1.6394	1.6493	1.6476	1.6348
	64	–	–	1.5741	1.5840	1.5662
3 of 8, per forward	16	1.7387	1.7351	1.7368	1.7335	1.7288
	32	–	1.6418	1.6376	1.6390	1.6348
	64	–	–	1.5709	1.5733	1.5662
3 of 8, per token	16	1.7302	1.7316	1.7408	1.7303	1.7288
	32	–	1.6402	1.6402	1.6337	1.6348
	64	–	–	1.5702	1.5728	1.5662
hard Top-$(K{+}J)$		1.6348	1.5662	1.5178	1.4840	

No cell is below the hard Top-K of its row by more than 0.0011 (3 of 8 per token, $(32, 224)$), and no cell is closer than 0.05 to the hard Top-$(K{+}J)$ of its column. Differences of 0.01 between cells are within what one seed resolves.

6 The noise does not die out

In the selected runs the share of the drawn support outside the deterministic Top-K grows during training, from 0.08–0.11 at step 200 to 0.20–0.39 at step 20000, more for larger J (Figure 5). The model does not escape the noise by enlarging the gaps between scores; the loss is nevertheless that of hard Top-K.

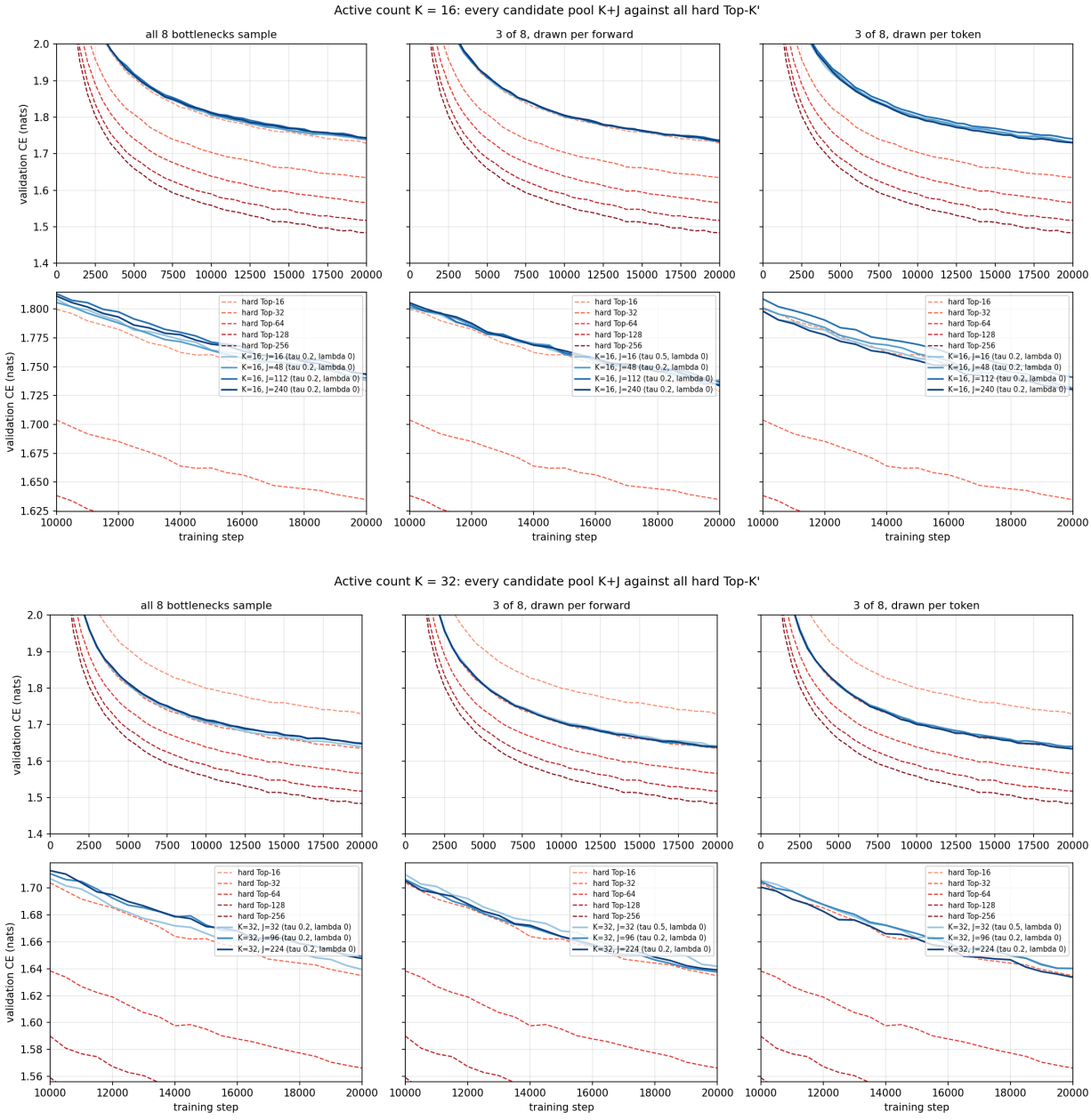


Figure 3: Per active count, $K = 16$ (top) and $K = 32$ (bottom): every candidate pool (solid, blues light to dark with J) against all hard Top-K' (dashed, reds light to dark with K'); one column per variant; top rows the whole run, bottom rows steps 10000–20000 between hard Top-$2K$ and hard Top-K. Every Gumbel curve runs along hard Top-K.

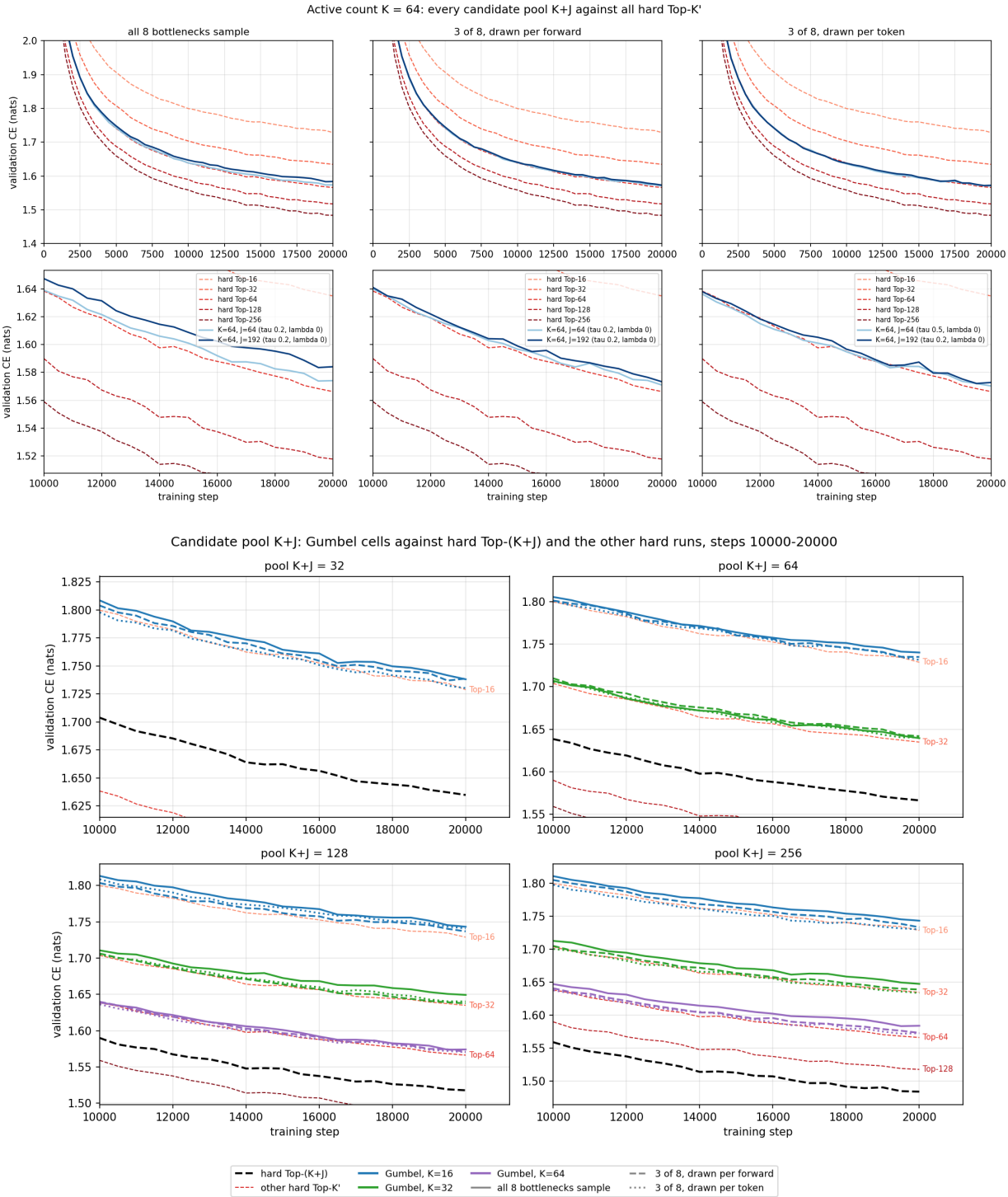


Figure 4: Top: $K = 64$, as in Figure 3. Bottom: per candidate pool $K+J$, the Gumbel cells with that pool (color: K; line style: variant) against hard Top-$(K+J)$ (black dashed) and the other hard runs. Each Gumbel cell lies on the hard curve of its own K, not near that of its pool.

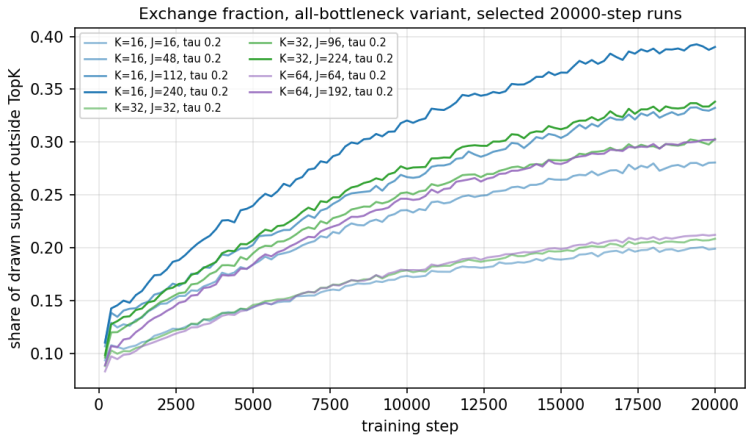


Figure 5: Share of the drawn support outside the deterministic Top-K during the selected 20000-step runs, all 8 bottlenecks sampling, $\tau = 0.2$, $\lambda = 0$.

7 The policy term over 20000 steps

The search compares settings at step 2500. To test whether a setting with the policy term overtakes hard Top-K later in training, the largest λ that trains, $\lambda = 10^{-4}$, was continued to step 20000 ($3 \cdot 10^{-4}$ stalled in the probe of the 3-of-8 variants, 10^{-3} collapsed by step 1000). It was continued at $\tau = 0.5$ and 1.2, the two temperatures with the lowest loss at this λ in all 27 cell–variant pairs; at $\tau = 0.2$ the validation CE at step 2500 is 2.49 to 4.72. These are 54 runs (9 cells, 3 variants, 2 temperatures), each resumed from its own step-2500 search checkpoint. Under the constant learning rate this is the run the search started, continued.

Divergence guard and replacements. A guard stops a run when its training CE stays more than 2.5 nats above its best value for 800 steps. Where it stopped a $\lambda = 10^{-4}$ run, the run with the same cell, variant and τ at $\lambda = 3 \cdot 10^{-5}$ was continued in the same way from its own search checkpoint (7 runs; `scripts/gcr_lpos_fallback.py`).

Collapse. A run is called collapsed when the guard stopped it, or when the share of its drawn support outside the deterministic Top-K, averaged over its last 500 steps, is below 0.02: the sampling has become deterministic. At the end of the runs that did not collapse this share is 0.07 to 0.76.

The quantity compared is the margin over hard Top-K at the same step,

$$\Delta(s) = \mathrm{CE}_{\mathrm{val}}(s) - \mathrm{CE}_{\mathrm{val}}^{\text{hard Top-}K}(s), \tag{3}$$

so that a crossover is $\Delta(s) < 0$.

Measurements: margin over hard Top-K

No $\lambda > 0$ run reaches hard Top-K at any validation step between 2500 and 20000 (Figures 6 and 7). The smallest margin over all 61 runs and all steps is +0.024 (3 of 8 per forward, $(32, 32)$, $\tau = 0.5$, $\lambda = 3 \cdot 10^{-5}$, step 16500). Mean margin of the runs that did not collapse:

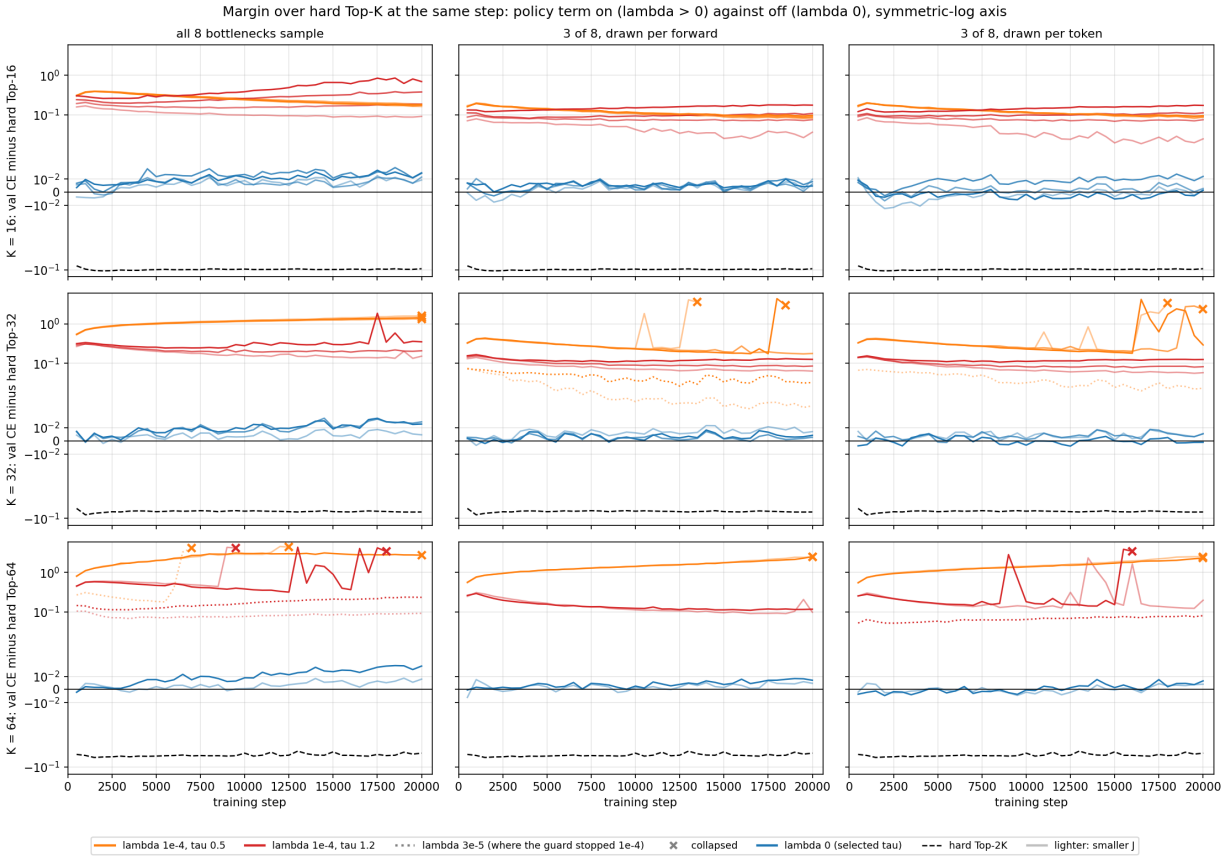


Figure 6: Margin $\Delta(s)$ over hard Top-K for every $\lambda > 0$ run past step 2500 (orange: $\lambda = 10^{-4}$, $\tau = 0.5$; red: $\lambda = 10^{-4}$, $\tau = 1.2$; dotted: $\lambda = 3{\cdot}10^{-5}$ at the same τ; lighter for smaller J; $\times$: collapsed), the selected $\lambda = 0$ runs (blue) and hard Top-$2K$ (black dashed). Symmetric-log axis, linear between ± 0.05.

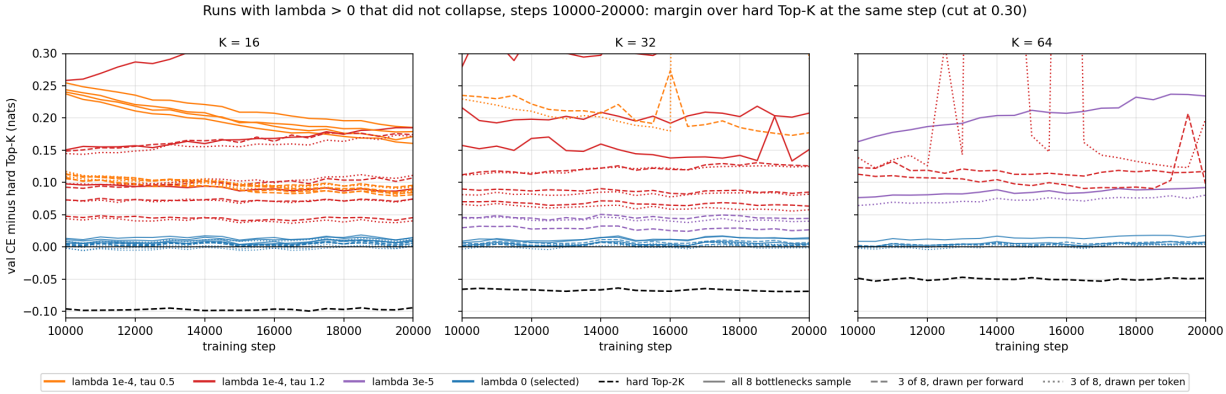


Figure 7: The $\lambda > 0$ runs that did not collapse, steps 10000–20000, linear axis cut at 0.30; color: setting ($\lambda = 3 \cdot 10^{-5}$ in purple), line style: variant; blue: selected $\lambda = 0$ runs; black dashed: hard Top-$2K$.

variant	setting	n	2500	10000	15000	20000	min. at 20000
all 8 sample	$\lambda = 10^{-4}$, $\tau = 0.5$	4	+0.375	+0.244	+0.196	+0.174	+0.161
	$\lambda = 10^{-4}$, $\tau = 1.2$	7	+0.226	+0.220	+0.266	+0.293	+0.090
	$\lambda = 3 \cdot 10^{-5}$	2	+0.092	+0.120	+0.148	+0.163	+0.092
	$\lambda = 0$ (selected)	9	+0.002	+0.008	+0.008	+0.012	+0.005
3 of 8, per forward	$\lambda = 10^{-4}$, $\tau = 0.5$	5	+0.206	+0.135	+0.113	+0.105	+0.081
	$\lambda = 10^{-4}$, $\tau = 1.2$	9	+0.127	+0.097	+0.094	+0.099	+0.046
	$\lambda = 3 \cdot 10^{-5}$	2	+0.055	+0.038	+0.035	+0.036	+0.027
	$\lambda = 0$ (selected)	9	−0.000	+0.003	+0.002	+0.006	+0.003
3 of 8, per token	$\lambda = 10^{-4}$, $\tau = 0.5$	5	+0.213	+0.137	+0.114	+0.130	+0.086
	$\lambda = 10^{-4}$, $\tau = 1.2$	8	+0.116	+0.095	+0.099	+0.108	+0.041
	$\lambda = 3 \cdot 10^{-5}$	2	+0.056	+0.054	+0.057	+0.060	+0.040
	$\lambda = 0$ (selected)	9	−0.002	+0.000	+0.001	+0.004	−0.001

The $\lambda > 0$ rows average different cells: the collapsed runs are left out, and $\lambda = 3 \cdot 10^{-5}$ was run only where the guard stopped $\lambda = 10^{-4}$. The five runs closest to hard Top-K at step 20000, with the least-squares slope of Δ over steps 15000–20000 in nats per 10000 steps:

variant, (K, J)	τ, λ	2500	5000	10000	15000	20000	slope
3 of 8 per forward, $(32, 32)$	0.5, $3 \cdot 10^{-5}$	+0.051	+0.042	+0.030	+0.026	+0.027	+0.001
3 of 8 per token, $(32, 32)$	0.5, $3 \cdot 10^{-5}$	+0.061	+0.052	+0.044	+0.041	+0.040	−0.001
3 of 8 per token, $(16, 16)$	1.2, 10^{-4}	+0.065	+0.056	+0.044	+0.038	+0.041	+0.003
3 of 8 per forward, $(32, 224)$	0.5, $3 \cdot 10^{-5}$	+0.059	+0.053	+0.046	+0.045	+0.044	−0.002
3 of 8 per forward, $(16, 16)$	1.2, 10^{-4}	+0.064	+0.061	+0.048	+0.041	+0.046	+0.007

- In these runs the margin shrinks by 0.017–0.025 between steps 2500 and 12500 and changes by −0.002 to +0.007 per 10000 steps after step 15000. Over all 44 runs that did not collapse, the median slope is −0.005 per 10000 steps over steps 10000–20000 and +0.005 over steps 15000–20000.
- The fastest decline is with all 8 bottlenecks sampling at $K = 16$, $\tau = 0.5$, $\lambda = 10^{-4}$: the margin falls from 0.24–0.25 at step 10000 to 0.19–0.21 at step 15000 and 0.16–0.19 at step 20000, so the decrease halves from the first to the second 5000 steps.
- With all 8 bottlenecks sampling at $\tau = 1.2$ the mean margin grows after step 5000; the runs with pools 128 and 256 at $K = 16$ rise by 0.05 and 0.26 from their lowest CE while the share outside Top-K is 0.67–0.76.
- Per K, the closest $\lambda > 0$ run ends +0.041 ($K = 16$), +0.027 ($K = 32$) and +0.081 ($K = 64$) above hard Top-K. The selected $\lambda = 0$ runs end between −0.001 and +0.018, and their mean margin grows by 0.006 to 0.010 from step 2500 to 20000.

Measurements: collapses

16 of the 54 runs at $\lambda = 10^{-4}$ collapse, and 1 of the 7 at $3 \cdot 10^{-5}$. They fall into two groups (table below).

- **Collapsed by step 2500** (9 runs, all $\tau = 0.5$, $\lambda = 10^{-4}$): the share outside Top-K is 0.003–0.021 at step 2500 and the margin 0.86–1.53. The CE does not recover, and rises to 2.97–4.32 by step 20000. The guard stopped one of these runs.
- **Trained, then collapsed** (7 at $\lambda = 10^{-4}$, 1 at $3 \cdot 10^{-5}$): these runs reach a lowest validation CE of 1.74 to 2.10, 0.14 to 0.44 above hard Top-K at the same step. The first jump of the training CE comes between steps 6400 and 18000, and the final collapse between steps 7460 and about 19000 (Figure 8).

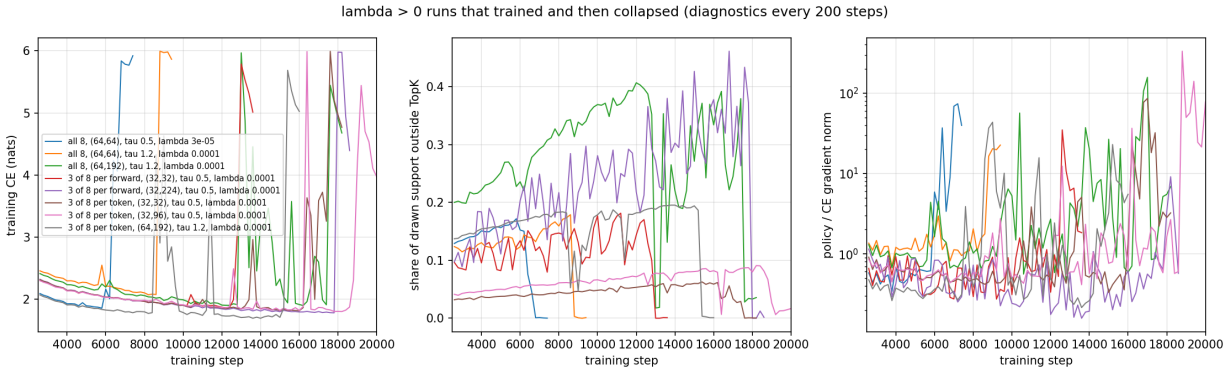


Figure 8: The 8 runs that trained and then collapsed: training CE, share of the drawn support outside Top-K, and the norm of the policy term's gradient relative to the CE gradient on the same micro-batch, from step 2500 (diagnostics every 200 steps).

  - The training CE jumps to 5–6 nats (the unigram level) within a few hundred steps. In several runs it recovers partly once or twice before the last jump.
  - At the jumps, the ratio of the policy gradient's norm to the CE gradient's reaches 9 to 330. Before the first jump the ratio is mostly between 0.2 and 2, with isolated values of 3.9 to 56.
  - After the last jump the share outside Top-K falls to 0.001–0.033.
  - Apart from those isolated ratio values, the logged diagnostics show no trend before the first jump.

- The guard missed 9 of the 17 collapses: 8 of the first group, whose CE rises slowly, and 3 of 8 per token at $(32, 96)$, whose final collapse began about 1000 steps before the end. Their $\lambda = 3 \cdot 10^{-5}$ counterparts were not run.

variant	(K, J)	τ	λ	CE at 2500	lowest after 2500	end
collapsed by step 2500						
all 8	$(32, 32)$	0.5	10^{-4}	2.861	2.835 (3500)	3.217 at 20000
all 8	$(32, 96)$	0.5	10^{-4}	2.819	2.796 (5000)	2.967 at 20000
all 8	$(32, 224)$	0.5	10^{-4}	2.839	2.819 (4500)	3.076 at 20000
all 8	$(64, 64)$	0.5	10^{-4}	3.420	3.420 (2500)	guard stop at 12580
all 8	$(64, 192)$	0.5	10^{-4}	3.402	3.402 (2500)	4.321 at 20000
3 of 8 per fwd.	$(64, 64)$	0.5	10^{-4}	2.847	2.847 (2500)	4.036 at 20000
3 of 8 per fwd.	$(64, 192)$	0.5	10^{-4}	2.837	2.837 (2500)	4.097 at 20000
3 of 8 per tok.	$(64, 64)$	0.5	10^{-4}	2.856	2.852 (3000)	4.080 at 20000
3 of 8 per tok.	$(64, 192)$	0.5	10^{-4}	2.829	2.816 (3500)	3.880 at 20000
trained, then collapsed						
all 8	$(64, 64)$	1.2	10^{-4}	2.484	2.101 (8500)	guard stop at 9540
all 8	$(64, 192)$	1.2	10^{-4}	2.449	1.931 (12500)	guard stop at 18240
all 8	$(64, 64)$	0.5	$3 \cdot 10^{-5}$	2.129	1.905 (5500)	guard stop at 7460
3 of 8 per fwd.	$(32, 32)$	0.5	10^{-4}	2.349	1.921 (12000)	guard stop at 13660
3 of 8 per fwd.	$(32, 224)$	0.5	10^{-4}	2.355	1.821 (17500)	guard stop at 18640
3 of 8 per tok.	$(32, 32)$	0.5	10^{-4}	2.356	1.851 (15500)	guard stop at 18360
3 of 8 per tok.	$(32, 96)$	0.5	10^{-4}	2.338	1.845 (18000)	3.992 at 20000
3 of 8 per tok.	$(64, 192)$	1.2	10^{-4}	2.107	1.739 (14000)	guard stop at 16100

Hard Top-K at step 2500: 2.064, 1.959 and 1.891 for $K = 16, 32, 64$. The 6 $\lambda = 3{\cdot}10^{-5}$ replacements that did not collapse end +0.027 to +0.234 above hard Top-K.

8 Interpretation

The simplest reading: with $\lambda = 0$ a Gumbel Top-$(K{+}J)$ bottleneck is hard Top-K with random exchanges near the K-th score. Exchanges between coordinates of similar magnitude cost nothing measurable, and the forward never uses more than K coordinates, so the model cannot gain what hard Top-$(K{+}J)$ gains. The policy gradient is the only term that reaches the candidates outside the drawn support, but its estimate is dominated by noise: its norm equals the CE gradient's near $\lambda = 10^{-4}$, where it already costs 0.3–1.5 nats at step 2500 (mean over cells, depending on τ), and at $\lambda = 10^{-5}$, a tenth of the CE gradient, it still costs 0.015–0.07 nats and improves no cell. Sampling at 3 of 8 bottlenecks lowers that noise by about the factor the gradient ratios show, not enough to change the outcome.

Continuing the policy term to step 20000 does not change this reading (Section 7). Where it trains, the margin over hard Top-K shrinks in the first half of training and is flat in the second. A linear extrapolation of the decline of the closest run over steps 10000–20000 (-0.005 nats per 10000 steps) would reach hard Top-K after about 57000 further steps; over steps 15000–20000 the same run's margin does not decline. The long runs also show a cost the search could not: of the 45 runs at $\lambda = 10^{-4}$ that trained normally, 7 collapse later, the last after more than 15000 steps beyond the search, each time with the policy gradient briefly 9 to 330 times the CE gradient. A policy term large enough to move the support makes the training unstable before it brings the loss below hard Top-K.

9 Scope

Single runs at seed 1337, on two boxes of the same GPU type; differences of 0.01 nats are within run-to-run variation. Constant learning rate, so the absolute values are not comparable with the cosine-schedule K+J notes. abs-TopK selection only. The (τ, λ) search ranks settings at step 2500. Section 7 continues $\lambda = 10^{-4}$ at $\tau = 0.5$ and 1.2 to step 20000, with $3 \cdot 10^{-5}$ where the guard stopped it; $\lambda = 10^{-5}$, $\lambda = 3 \cdot 10^{-5}$ elsewhere and $\tau = 0.2$ with $\lambda > 0$ were not continued, and the 9 collapses the guard missed were not replaced. The runs end at step 20000, so an advantage appearing only after that is not excluded. The collapse threshold (share outside Top-K below 0.02 over the last 500 steps) is a choice made here; every collapsed run is listed.

Data. Runs on Drive `runs_ankara` (all 8; 3 of 8 per forward; hard Top-K) and `runs_bern` (3 of 8 per token; the remaining all-8 cells); selections in `analysis_{ankara,bern}/gcr_selection_*.json` and `docs/figures/gumbel-cr/data/`; figures and tables: `scripts/plot_gumbel_cr_report.py`. The $\lambda > 0$ continuation resumed the search runs in place (same run directories; the console log of the search part kept as `<name>.search.log`); its replacement decisions are in `docs/figures/gumbel-cr/data/lpos_fallback_{ankara,bern}.json` (`scripts/gcr_lpos_fallback.py`).